

## EXPERIMENT - 2

TO PREPARE THE SATURATED SOLUTION OF IMPURE SALT AND OBTAIN THE PURE CRYSTAL OF THE SAME SALT BY CRYSTALLIZATION PROCESS.

### APPARATUS REQUIRED

- |                                   |                    |
|-----------------------------------|--------------------|
| a) Beaker                         | b) Porcelain basin |
| c) Test tubes and test tube stand | d) Filter paper    |
| e) Tripod stand                   | f) Bunsen burner   |
| g) Sand bath                      | h) Wire gauze      |
| i) Wash bottle                    |                    |

### CHEMICAL REQUIRED

- i. Bazaar or Impure Copper Sulphate

### THEORY

Crystallization is the process of making crystal by heating saturated solution of a given substance up to its crystallization point followed by cooling process.

The mother liquor (or spent liquor) is the solution remaining after a component has been removed by crystallization.

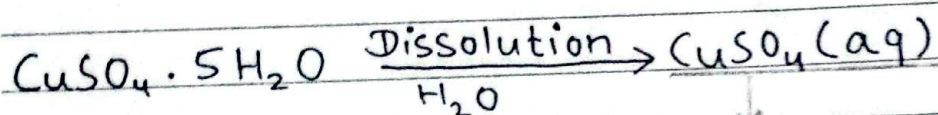
The solution which can dissolve no more solute at a given temperature is called saturated solution. \* Unsaturated, Supersaturated solution.

Solubility is defined as the maximum amount

of a substance that will dissolve in a given amount of solvent at a specified temperature.

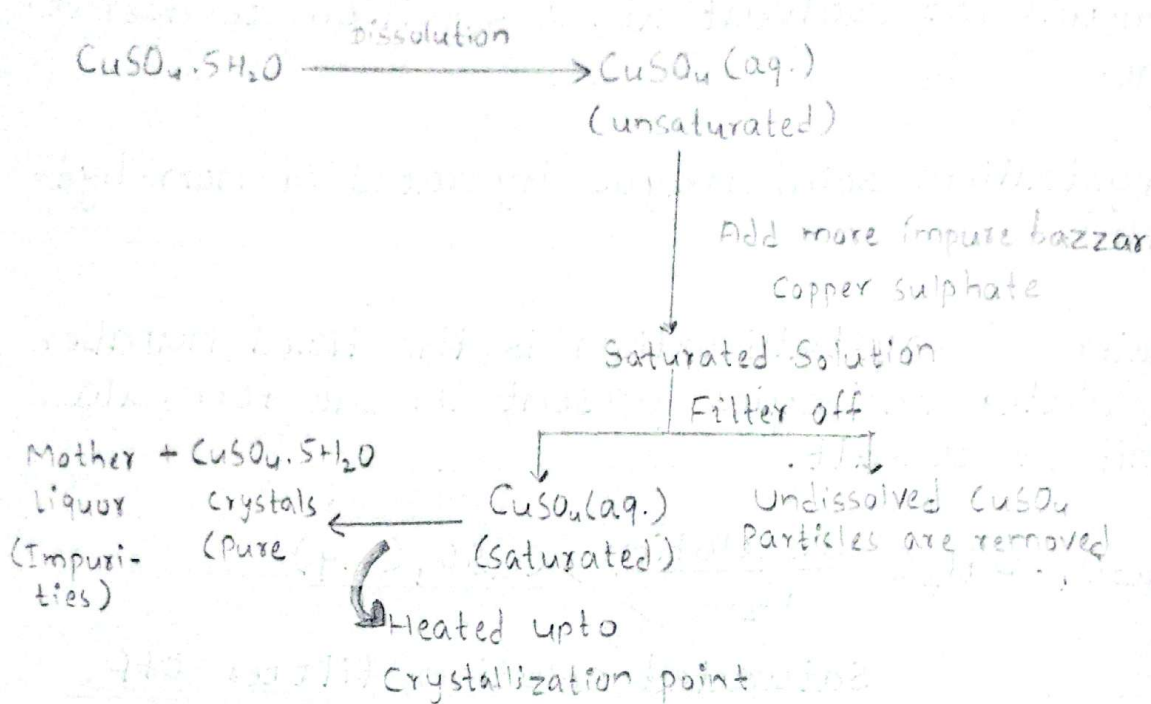
Crystalline solid maybe hydrated or non-hydrated.

Water of crystallization is the fixed number of water molecules present in one formula unit of a salt.



saturated solution filter off.



Outline of the process

Copper sulphate solution

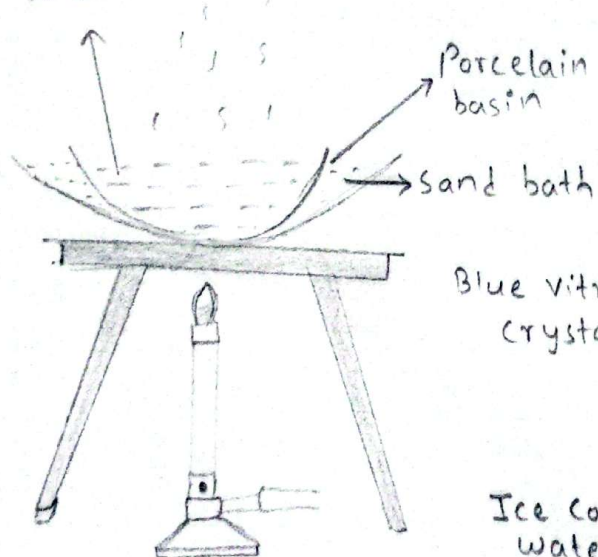


fig- evaporation

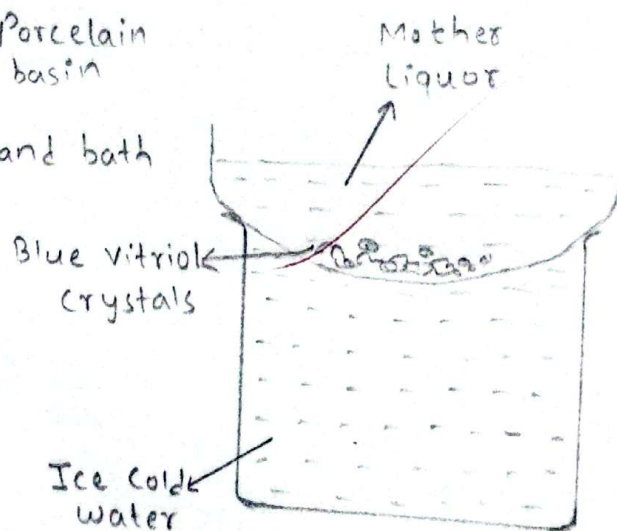


fig- Crystallization

**PROCEDURE**

About 15 ml of water was taken in a beaker, and impure copper sulphate was added little by little with constant stirring until a saturated solution was formed. The saturated solution containing some undissolved crystals was then ~~fixed~~ filtered, and the filtrate was collected in a porcelain basin. The solution was evaporated with occasional stirring over a sand bath until the crystallization point was reached. The crystallization point was checked by transferring a drop of the hot solution onto the outer wall of a test tube containing cold water with the help of a glass rod. When tiny needle-shaped crystals appeared on the wall of the test tube, heating was stopped. The porcelain basin was taken out with the help of tongs and placed to float on cold water in a water trough. After cooling, blue crystals began to form and continued forming for some time. When crystallization was completed, the remaining liquid, known as the mother liquor, was separated by decantation. The pure crystals were collected on filter paper, carefully spread without breaking their shape, and left to dry. Finally, the dried crystals were submitted to the lab supervisor for inspection.



OBSERVATION TABLE (Test of crystallization point of copper sulphate)

| S.N.                                       | Experiments                                                                                                                               | Observations              | Inferences                                  |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------|
| 1.                                         | Few drops of hot saturated solution were transferred on the wall of a test tube containing cold water with the help of a glass rod.       | No tiny crystals appeared | Crystallization point (C.P) was not reached |
| After heating the solution for few minutes |                                                                                                                                           |                           |                                             |
| 2.                                         | Few drops of hot saturated solution were again transferred on the wall of a test tube containing cold water with the help of a glass rod. | Tiny crystals appeared    | Crystallization point (C.P) was reached     |

**RESULT**

In this way pure crystals of blue vitriol i.e. copper sulphate pentahydrate were obtained from bazaar copper sulphate and submitted to supervisor for inspection.

**CONCLUSION**

Hence from above experiment it can be concluded that pure crystals can be obtained by heating saturated solution at temperature upto crystallization point followed by cooling process.



**PRECAUTIONS**

1. Personal protective equipment should be worn.
2. Saturated solution must be prepared at laboratory temp. without heating content in the beaker.
3. Crystallization point should not be exceeded.
4. Mother liquor containing soluble impurities should be discarded carefully.
5. Cooling should be done slowly to get good crystals.
6. Crystals formed should not be heated over Bunsen flame otherwise crystals lose all water of crystallization and converted into anhydrous form of amorphous copper sulphate.

- VIVA-VOCE**
1. What is the molecular formula and systematic name of blue vitriol?
  2. What do you mean by water of crystallization?
  3. What are hygroscopic solids?

1. Molecular formula -  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Systematic name - Copper(II) sulphate pentahydrate

2. Water of crystallization is the fixed number of water molecules present in a crystal that give it shape and color.

3. Solids that are capable of absorbing moisture (water molecules) when exposed to open atmosphere are called hygroscopic solids.

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